REJECTED DRAFT — FOR REVIEW ONLY

This draft failed the post-polish native audit after its single permitted repair. Nine sentences were flagged for unsupported elaboration or literature positioning outside the permitted section. It is not an accepted publication. The delivered fallback retains the original draft. The underlying article is 4,658 reading words, excluding references and this notice.

Order, Consistency and Information States: A Modest Adequacy Condition for the Semantics of Epistemic Modals

Abstract

Dynamic semantics gives conjunction a sequential interpretation and treats the might-operator as a consistency test. It also records an order contrast: once the information that it is not raining has been added, the information that it might be raining is no longer consistent with it, whereas in the opposite order there is no problem. This paper argues that these commitments and this contrast jointly support one adequacy condition: an adequate semantics of epistemic modals must make modal evaluation sensitive to information states, not only to worlds. The argument runs from sign to conclusion. Because the might-test is run against an information state, and sequential addition changes that state, the order contrast counts as evidence that evaluation depends on information states. The paper takes up two critical questions, whether the contrast is reliable and whether the commitments support the conclusion, and answers both affirmatively within stated limits. The conclusion is conditional on its premises, leaves worlds a role, and does not show that every epistemic modal behaves this way. A bounded literature search indicates that the condition is partially anticipated by existing work, so the contribution is modest: a clear statement of one supported inference and its scope.

1. Introduction

This paper defends a single adequacy condition on the semantics of epistemic modals: an adequate semantics of epistemic modals must make modal evaluation sensitive to information states, not only to worlds. The condition is something any adequate account has to meet. It is not a report that some account has already been independently shown to meet it, and it is not a rival semantics. Its wording also matters. The condition rules out evaluation based on worlds alone; it does not say that worlds play no part in modal evaluation. What it requires is that information states also figure in how epistemic-modal claims are evaluated. The rest of the paper explains why the cited material supports this requirement, how strong that support is, and where it stops. Throughout, the aim is to state one inference clearly and to keep its conclusion at the size the evidence allows, rather than to settle the wider debate about how epistemic modals should be modelled.

The condition rests on two grounds, both taken from the dynamic-semantic tradition and endorsed in the cited source. The first is a pair of semantic commitments: dynamic semantics gives conjunction a sequential interpretation and interprets the might-operator as a consistency test (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). The second is an order contrast in discourse. After the information that it is not raining has been added, that information is no longer consistent with the information that it might be raining; in the opposite order there is no problem, so order matters in discourse (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). These two grounds are the premises of the argument. They are not further conclusions about every epistemic modal, and the argument does not try to establish them independently. Its task is narrower: to show how, granted these premises, the contrast bears on modal evaluation and why it supports the adequacy condition rather than something stronger or weaker.

The contribution is modest, and I state this plainly at the outset rather than hedging each later claim. The argument is conditional: if the commitments hold and the contrast is as described, then the contrast supports the requirement. Support runs one way, from the commitments and the order effect to the requirement on an adequate account; the requirement is not used to vindicate the commitments. The central example also has a limited reach. It supports the claim that information states must matter to modal evaluation, but it does not by itself show that every epistemic modal behaves as the example does. Finally, as Section 2 records, a bounded search of recent literature indicates that the requirement is partially anticipated by existing work. What the paper offers is therefore a careful reconstruction of one supported inference, its two critical questions and its limits, not a new theory and not a certified novelty claim.

The paper proceeds as follows. Section 2 locates the condition relative to recent literature and states the scope of any novelty claim. Section 3 sets out the argument. It first explains each of the two semantic commitments and how they combine, then presents the order contrast and fixes its scope, and finally traces the inference from these premises to the adequacy condition. Section 4 develops three objections, each grounded in the critical questions the argument must answer: whether the order contrast is reliable, whether the semantic commitments really support the conclusion, and whether the conclusion overreaches. Section 5 collects the substantive limits of the support in one place, so that they need not be repeated as caveats elsewhere. Section 6 concludes by stating what the argument establishes and what it leaves open. Each section has a single argumentative job, and the thesis is not restated except where the inference requires it.

2. Related work and scope of the novelty claim

The literature context for this paper comes from a bounded search, not an exhaustive review. The search covered 2018–2026 literature on contextualism, relativism, expressivism, dynamic semantics, epistemic contradictions, probability operators, retraction and assessment sensitivity, with a window from 1 January 2018 to 28 October 2026 (Literature-currency search 2026, 2018–2026 search). It was a ranked search of indexes and public sources. It obtained a number of recent full texts, including recent surveys, formal models, empirical tests and responses, but several selected recent works and older originals remained unavailable, and older foundational chapters, complete books and recent

empirical reviews are recorded as material access gaps (Literature-currency search 2026, 2018–2026 search). The search record itself says that a failed search or an inaccessible paper does not establish an open problem or novelty (Literature-currency search 2026, 2018–2026 search). Nothing below should therefore be read as a claim that the adequacy condition is new, or that no prior work states it. The section records only what the read works show about overlap with the paper's thesis and how far that overlap extends.

On the thesis itself, the search supports partial anticipation. The condition that an adequate semantics must make modal evaluation sensitive to information states, not only to worlds, is partially anticipated by a 2025 survey of dynamic semantics for epistemic modality, which surveys dynamic and static options (PDF pp. 1, 2, 4, 12, 16, 17; §§2–6) (Mandelkern 2025, PDF pp. 1, 2, 4, 12, 16, 17; §2–6, especially6.2 Order). The version read is an author chapter manuscript of excerpts, not the full book. Work giving a pragmatic competition account of modal embedding contrasts bears on the condition differently: it holds that modal embedding is not automatically evidence uniquely favouring one semantic family (PDF pp. 1, 2), and it figures in the search's assessment as one of the alternatives rather than as an anticipation of the condition (Močnik 2019, PDF pp. 1, 2; §abstract and introduction). That assessment is that information-sensitive approaches exist, but that alternatives challenge whether the relevant data require a separate semantic parameter or a nonclassical semantics; this paper does not resolve that challenge (Mandelkern 2025, PDF pp. 1, 2, 4, 12, 16, 17; §2–6, especially6.2 Order) (Močnik 2019, PDF pp. 1, 2; §abstract and introduction). A 2022 paper defending classical logic by treating epistemic contradictions as epistemically defective rather than semantically inconsistent bears on this question, but only its abstract was available, so it is named here as an access gap, not as a read text. A critical notice directly comparing dynamic semantics with dynamic propositionalism was read (PDF pp. 1, 2), but no neutral resolution of that debate is inferred from it (Willer 2023, PDF pp. 1, 2; §The Plot; critical notice of Stojnić).

Two nearby claims should be separated from the paper's thesis. First, order sensitivity in epistemic-modal discourse as a diagnostic constraint on an adequate semantics is already addressed in the surveyed literature. The same survey explicitly limits that diagnostic: discourse-order effects do not by themselves establish within-sentence semantic order effects, and its claims about Boolean fragments are restricted (especially §6.2, Order) (Mandelkern 2025, PDF pp. 1, 2, 4, 12, 16, 17; §2–6, especially6.2 Order). Using the order contrast as a diagnostic is therefore not presented here as a contribution. Second, the stronger claim that a static possible-worlds semantics treating sentence sequences as conjunctions cannot explain order-sensitive discourse is only partially addressed. On the read evidence, rejecting all static semantics would require addressing pragmatics and embedded symmetry (Mandelkern 2025, PDF pp. 1, 2, 4, 12, 16, 17; §2–6, especially6.2 Order) (Močnik 2019, PDF pp. 1, 2; §abstract and introduction). This paper does not make that stronger claim. Its scope is the adequacy condition, supported by the cited commitments and contrast, and these literature findings serve only to place the paper. They are not premises of the argument that follows.

3. Argument

The argument has the form of an argument from sign. Its conclusion is the adequacy condition. Its premises are the pair of semantic commitments and the order contrast. The contrast is the sign: an observable difference between two orders of adding information in discourse. The commitments explain why that sign bears on the conclusion. They fix how conjunction and might are interpreted, and under that interpretation the contrast becomes evidence about what modal evaluation depends on. The argument map below records this structure and the recorded answers to the two critical questions that test the link from premises to conclusion. The discussion takes the commitments first, then the contrast, then the inference. Keeping these steps apart matters, because each premise contributes something specific and limited. Neither premise alone yields the conclusion. The commitments without the contrast give no evidence that order affects evaluation, and the contrast without the commitments does not say why the difference between orders should concern information states.

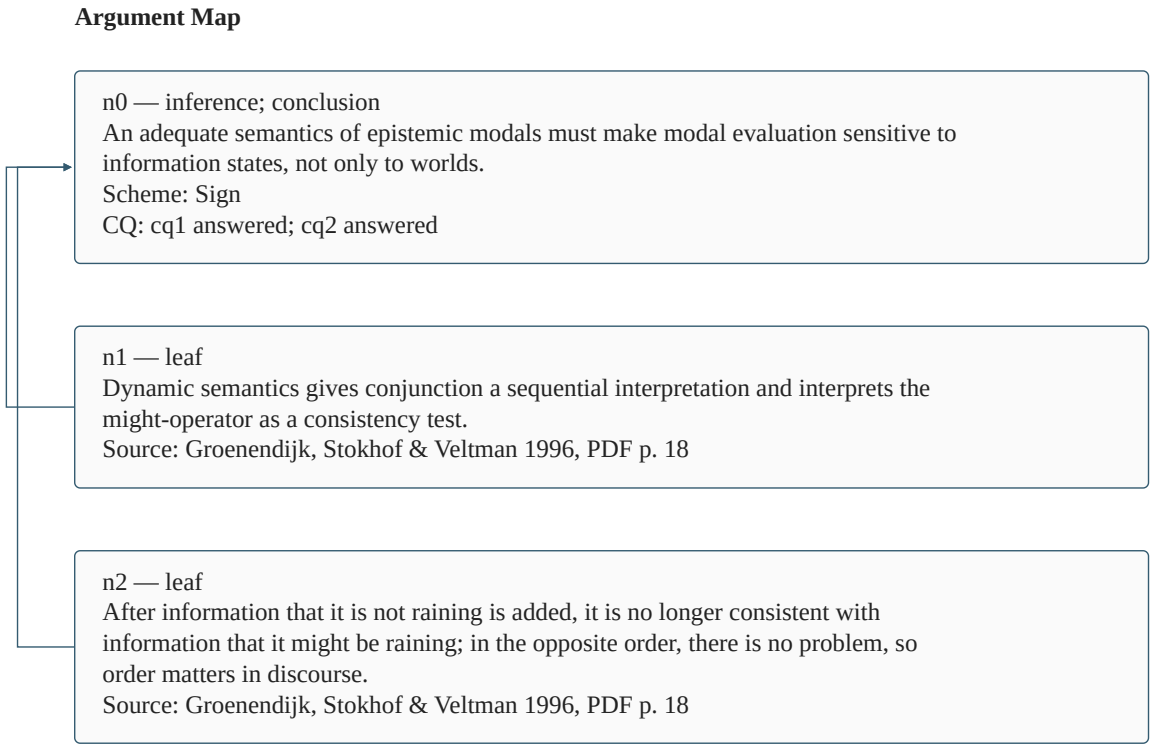


Figure 1. Route premises, inference schemes and conclusion. Arrows follow premise-to-conclusion links; CQ labels report recorded answers.

The first commitment concerns conjunction. Dynamic semantics interprets conjunction sequentially, rather than in a way that leaves out its sequential character (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). This fixes how the account treats conjunction, and that is its whole content. By itself it says nothing further about the consequences of any particular ordering. It also does not set out stages, outcomes or rules for how any particular conjunct changes an information state. It supplies only the sequential part of the proposal: on this treatment, a conjunction is processed in order, so that what

comes first is in place when what comes second is interpreted. This is what the argument needs from the first commitment. Because conjunction is interpreted sequentially, the order in which information arrives becomes part of what the semantics represents, and a contrast between orders can register in it.

The second commitment concerns the might-operator, which dynamic semantics interprets as a consistency test (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). The commitment does not say how the test works in detail, and it does not assign might any further function. What it does is tie modal evaluation explicitly to information. Under this interpretation, the question that bears on a might-claim is whether it remains consistent with the information state. A test of consistency is always a test against something, and here that something is a body of information rather than a single world. This is the point at which information states enter the account. The two commitments therefore contribute different things: the sequential treatment of conjunction concerns the order in which information is added, while the consistency-test treatment of might concerns whether a might-claim remains consistent with the resulting information state.

The two commitments concern different expressions, and their roles should be kept apart. Sequence belongs to the interpretation of conjunction; consistency with information belongs to the interpretation of might. Keeping the roles distinct shows what each contributes without adding a further semantic rule or giving either feature to the other expression. Taken together, they shape how the account treats epistemic-modal claims that appear in sequence: information is added in order, and might is checked for consistency against the information already in place. This limited pairing is what frames the evaluation of epistemic-modal claims against information states. Its scope is narrow. As stated, the commitments make no broader claim about all epistemic modals or other modal operators, about other properties of conjunction, or about consequences beyond the two commitments themselves. They are endorsed by the source as features of the dynamic account, and the argument uses them as such. It does not defend them further here.

The second premise is the order contrast. It concerns the sequence in which information is added in discourse. In one order, the information that it is not raining comes first and the information that it might be raining follows: "It isn't raining outside. … It might be raining outside." Once the not-raining information has been added, it is no longer consistent with the information that it might be raining. In the opposite order, there is no problem. From this contrast the source concludes that order matters in discourse (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). The example is simple, and its simplicity is useful. Only one thing varies between the two cases, and the result differs between them. This gives the contrast its value as a sign: an observable difference that can be traced to a single feature of the discourse, the order of addition, and not to any difference in the information being added.

The contrast lies in the order, not in the two pieces of information taken apart from it. The same not-raining and might-be-raining information appears in both orders; only the order of addition differs. The consistency failure belongs to the not-raining-first order. The claim about the reverse order says only that it presents no problem, and it gives no further result for that sequence. "Order matters" names this difference between the two orders. It does not add a further outcome or a general rule

(Groenendijk, Stokhof & Veltman 1996, PDF p. 18). The contrast therefore concerns consistency following the addition of information, rather than an order-free description of the two items, and so it concerns the state that results from addition.

The scope of the example should be stated once and then kept. It concerns these two pieces of information in these two orders. It says nothing about every possible order, every kind of information, every epistemic modal or any other example, and the argument draws no conclusion about other sequences or other modal expressions (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). Within that scope, the contrast is the premise the argument uses. It supports the requirement that information states matter to modal evaluation, but on its own it does not show that every epistemic modal behaves in this way. The argument relies on the example only for the narrower point that, in this discourse, whether the might-information is consistent depends on what has already been added.

The inference connects the premises through the role of the consistency test. If might tests consistency on an information state and conjunction adds information sequentially, then adding information in a different sequence changes whether the might-claim remains consistent. That is just what the contrast records. When the not-raining information comes first, the information state against which the might-claim is tested already contains it, and the might-information is no longer consistent with that state. When the order is reversed, the might-claim is tested before the not-raining information is in place, and there is no problem. The order effect is therefore explained by the way the consistency test applies to information added in sequence (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). On that basis, the contrast counts as a sign that modal evaluation depends on information states. The difference between the two orders is a difference in the state against which the test is run, and the might-test is a test of consistency with that state.

This is why the support goes to information-state-sensitive evaluation, not to evaluation by worlds alone. Because the might-test depends on consistency with an information state, and the sequence of addition changes that state, the contrast supports information-state-sensitive evaluation rather than evaluation by worlds alone (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). The direction of support should be explicit. The argument moves from the commitments and the specified order effect to a requirement on an adequate semantics. It does not move back from the requirement to the commitments, and the requirement is not offered as an independent proof that conjunction is sequential or that might tests consistency. Nor does the inference claim that the contrast could be explained in only one way. Its claim is that, given the stated commitments, the contrast is explained by the dependence of the might-test on an information state. That makes it evidence that an adequate semantics must give information states a role in modal evaluation. The source-support figure below shows how each node in this inference is backed by the frozen evidence.

Source Support

n0 ← Groenendijk, Stokhof & Veltman 1996, PDF p. 18
Stance: endorsed; attributed to Groenendijk, Stokhof and Veltman; role: cq_evidence

n0 ← Groenendijk, Stokhof & Veltman 1996, PDF p. 18
Stance: endorsed; attributed to Groenendijk, Stokhof and Veltman; role: cq_evidence

n1 ← Groenendijk, Stokhof & Veltman 1996, PDF p. 18
Stance: endorsed; attributed to Groenendijk, Stokhof and Veltman; role: global_premise

n2 ← Groenendijk, Stokhof & Veltman 1996, PDF p. 18
Stance: endorsed; attributed to Groenendijk, Stokhof and Veltman; role: global_premise

Figure 2. Frozen source support by route node, page citation, attribution and stance. Recorded support is not independent verification.

The conclusion reached has a definite shape. First, it is a requirement on an adequate semantics. It states what such a semantics must do, namely make modal evaluation sensitive to information states, and it does not claim that any particular semantics already succeeds. Second, it keeps a role for worlds. The requirement is that information states must also matter, not that they replace worlds as the basis of evaluation. The conclusion's wording, "not only to worlds", excludes world-only evaluation; it does not say that worlds have no role, and the argument offers nothing on that further question. Third, the requirement is supported by these commitments applied to this contrast. Its strength is the strength of that support, and the next section tests that support against the two critical questions the argument must answer.

4. Objections and replies

The first objection targets the sign. The argument treats the order contrast as reliable evidence about modal evaluation, but one might doubt that a single pair of discourses can carry this weight. The contrast involves one example about rain, two orders and one modal. Perhaps it records something about these particular sentences rather than anything about how epistemic-modal claims are evaluated. If the contrast is unreliable as a sign, the argument from sign fails: even granting the semantic commitments, there would be no evidence that order affects the evaluation of might-claims in the way the inference requires. The objection is a fair test, because the argument rests heavily on the example. The commitments explain why order could matter; the contrast is the only premise that shows that it does. So the reliability of the contrast as evidence, and not merely its truth as a description, is what is at issue.

The reply is that the contrast is reliable for the purpose the argument assigns it: as evidence that epistemic-modal evaluation is sensitive to information states. The order effect is explained when conjunction updates information sequentially and might tests consistency (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). Its reliability does not rest on the example being typical of all modal discourse. It rests on the fact that the two cases differ only in the order of addition, and that the difference in outcome is explained by the commitments. The same information appears in both orders, so the contrast cannot be put down to a difference in content. And because the explanation goes through the consistency test applied to sequentially added information, the contrast bears directly on what modal evaluation depends on. The reply comes with a limit that it does not hide. The contrast supports the claim that an adequate semantics must be sensitive to information states, but it does not by itself establish that every epistemic modal behaves this way. The objection is answered for the argument's conclusion, not for a universal generalization the argument does not make.

The second objection targets the link from the commitments to the conclusion. Grant that the contrast is reliable. One might still ask whether the semantic commitments actually support the adequacy condition, or whether they only describe one framework's way of handling the case. On this worry, the commitments say how dynamic semantics treats conjunction and might, but an adequacy condition is a claim about any adequate semantics. Moving from features of one framework to a demand on all adequate accounts may look like a step the premises do not license. Put differently, the commitments might seem to build information-state sensitivity into the description of the case, so that the conclusion is assumed rather than supported. This objection presses the second critical question directly: whether the commitments support the conclusion at all.

The reply turns on what the commitments do in the argument. They do support the conclusion: the might-operator is treated as a consistency test on an information state, and adding information in a different sequence changes whether the might-claim remains consistent. This supports information-state-sensitive evaluation rather than evaluation by worlds alone (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). The conclusion is not simply assumed. The commitments supply an explanation; the contrast supplies an observation the explanation fits; and the conclusion is drawn from the fit. On the framework worry, however, the reply concedes rather than overcomes. The support is conditional on the commitments, which are endorsed by the source and not independently established here. The argument claims that these commitments, applied to this contrast, support the requirement. It does not claim that the requirement could be derived without them, and it does not show that accounts rejecting the commitments are thereby bound by it.

The third objection is that the conclusion overreaches in one of two directions. It might be read as a claim about every epistemic modal, generalizing from one example to the whole class. Or it might be read as eliminating worlds from modal evaluation altogether and replacing world-based evaluation with information-state evaluation. On either reading, the conclusion would say more than the commitments and the contrast support. The first reading would need evidence about many modals, sequences and kinds of information, and the argument offers one well-described case. The second reading would need an argument that worlds play no role, and nothing in the premises addresses that

question. This objection does not dispute the inference traced in Section 3. It disputes the size of what the inference delivers, so the reply has to show that the conclusion, properly stated, stays within its support.

The reply is that the conclusion, as stated, makes neither claim. On generality, the example supports the requirement that information states must matter to modal evaluation, but it does not by itself establish that every epistemic modal behaves this way, and the conclusion does not need that wider claim (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). The conclusion is a requirement on an adequate semantics, not a generalization over the class of epistemic modals. On worlds, the conclusion says that modal evaluation must be sensitive to information states, not only to worlds. That is a demand that information states also figure in evaluation, not a demand that worlds be dropped. The objection is therefore correct about what the premises cannot support, and the conclusion already respects that. What would be mistaken is to recast the requirement as a universal claim about epistemic modals, or as an independently established fact. The argument offers neither.

5. Limits

The first limit concerns the status of the premises and the inference. The support is conditional. If conjunction is interpreted sequentially, might tests consistency, and the described order contrast holds, then the contrast supports the demand that an adequate semantics make modal evaluation sensitive to information states, not only to worlds (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). The premises are not independently established here. They are taken as endorsed commitments of the cited source. The contrast is also taken as stated. The source's own derivation of the contrast, which draws on further formal material, falls outside the supplied support and is not reconstructed. The argument therefore does not show that the contrast follows from the commitments by a particular formal route. It shows that, given the commitments, the contrast is explained by the dependence of the might-test on an information state, and that this explanation supports the adequacy condition. Readers who reject the commitments will not be moved by the argument as presented.

The second limit concerns reach. The contrast concerns the specified sequences: the not-raining information and the might-be-raining information, added in one order and then in the other. It is not a claim about every possible order, and nothing here supports a conclusion about other sequences, other kinds of information or other modalities (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). The particular order effect also does not by itself establish that every epistemic modal behaves this way. These limits do not weaken the conclusion as stated, because its content is only that information states must matter to the evaluation of epistemic modals, not only worlds. They do mean that the example counts as evidence for information-state sensitivity, and that this is all the support licenses. Any stronger thesis, whether about the behaviour of all epistemic modals, about the exclusion of worlds or about the inadequacy of particular rival frameworks, would need support that this paper does not supply.

The third limit concerns the evidential standing of the argument's own components. The answers to the two critical questions are recorded answers grounded in the cited commitments and contrast. They show that each question can be answered affirmatively on that support, but they are not independent verification of the premises, and the source-support figure should be read in the same way. The argument also uses only two cited items, both drawn from a single source, and both are used as premises rather than tested against further data. This is a deliberate restriction. It keeps the inference transparent, but it also means that the paper's conclusion is only as secure as those two items. Finally, the paper's contribution is modest in a further sense, already recorded in Section 2: the adequacy condition is partially anticipated by existing work. What is offered is a clear statement of one supported inference, its critical questions and its scope, not a new semantic proposal.

6. Conclusion

The paper has argued that the dynamic-semantic commitments and the not-raining order contrast together support an adequacy condition: an adequate semantics of epistemic modals must make modal evaluation sensitive to information states, not only to worlds. The argument is an argument from sign. Sequential conjunction makes the order of addition part of what the semantics represents. The consistency-test interpretation of might makes that order relevant to whether a might-claim passes. The contrast between the two orders, which contain the same information, is explained by the change in the state against which the test is run (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). Both critical questions received affirmative answers. The contrast is reliable as evidence of information-state sensitivity, and the commitments do support the conclusion, because might is tested for consistency against an information state that sequential addition changes.

What the argument does not establish matters as much as what it does. It does not show that every epistemic modal behaves as the example does. It does not say that worlds have no role in modal evaluation. It does not establish its premises independently, or reconstruct the source's own derivation of the contrast. And it does not claim novelty for the adequacy condition, which recent work partially anticipates. The result is a narrow but definite one: given these commitments and this contrast, information states must figure in the evaluation of epistemic modals. Any further question remains open as far as this argument goes. That includes which semantics best meets the condition, and whether the condition demands a separate semantic parameter or could be met in some other way.

References

Groenendijk, Jeroen; Stokhof, Martin; Veltman, Frank (1996). Coreference and Modality. In S. Lappin (ed.), The Handbook of Contemporary Semantic Theory, Blackwell. pp. 179-213.

Matthew Mandelkern (2025). Dynamic semantics for epistemic modality.

Maša Močnik (2019). Embedded Epistemic Modals Pragmatically. https://doi.org/10.18148/sub/2019.v23i2.606.

Malte Willer (2023). Dynamic semantics versus dynamic propositionalism. https://doi.org/10.1080/0020174x.2023.2174735.

Literature-currency record (2026). Bounded 2018–2026 search; verified page/section references and access limits retained in the accompanying publication receipt.